

TbUtilPkg Quick Reference

1. ZeroOneHot, OneHot

```
signal SA, SB, SC, SD : std_logic;
signal ZOH, OH : Boolean;
```

ZeroOneHot returns true when the std_logic_vector input either has a single one or is all zero.

If ZeroOneHot((SA & SB & SC & SD)) then

OneHot returns true when the std_logic_vector input either has a single one.

if OneHot((SA & SB & SC & SD)) then

2. Transaction Handshaking

Used for handshaking between the client side of a transaction interface and the model.

2.1 RequestTransaction

Request a transaction from the client side (TestCtrl).

```
procedure DoTransaction(
  ModelRec : inout ModelRecType ;
  DataIn   : in DataType
) is
begin
  ModelRec.Data <= DataIn ;
  RequestTransaction(
    Rdy => ModelRec.Rdy,
    Ack => ModelRec.Ack ) ;
end procedure DoTransaction ;
```

2.2 WaitForTransaction

Model side control to wait for client side (TestCtrl) to request a transaction

```
ExecuteTransProc : process
begin
  WaitForTransaction(
    Clk => Clk,
    Rdy => ModelRec.Rdy,
    Ack => ModelRec.Ack
  ) ;
  -- decode and execute transaction
```

3. Process to Process Synchronization

One process toggles a signal using the toggle procedure, the other process waits until the signal changes.

3.1 Toggle

Toggle a signal between 0 and 1. Signal type can be either bit or std_logic.

```
Toggle(Sync1) ;
Toggle(Sync2, 2 * Tperiod_clk) ;
```

3.2 WaitForToggle

Wait until a signal changes. Signal type can be either bit or std_logic.

```
WaitForToggle(Sync1) ;
```

4. WaitForBarrier

All processes stop until all processes have reached the barrier and called WaitForBarrier.

```
WaitForBarrier(Sync1) ;
WaitForBarrier(Sync2, 25 ms) ;
```

Sync1 may be either std_logic or BarrierType.

```
signal sync1 : std_logic ;
signal sync2 : BarrierType ;
```

5. WakeOnBarrier

Stop until the barrier is released. Does not put a driver on the barrier signal. Instead waits for all the other processes to reach the barrier, and then releases one delta cycle before WaitForBarrier. Allows a VC to wakeup as the test finishes and post completion status before the test ends.

```
WakeOnBarrier(TestDone) ;
```

6. Waiting for Clock

Wait for clock periods specified in either time units or an integral number of clock cycles. Is aligned to clock when it finishes.

```
WaitForClock(Clk, 5 * Tperiod_clk) ;
WaitForClock(Clk, 5) ;
```

7. Wait for Level

Wait until a signal is at a level.

```
WaitForLevel(A, '1') ; -- A='1'
WaitForLevel(Bool) ; -- TRUE
```